Radiometer: light, heat and molecular forces

A visible rotor connects heat transfer, rarefied gas dynamics and video measurement.

Measure how illumination and thermal history change rotation.

MIE 402 · Lecture 07 · Prof. Ehsan Roohi · Fall 2026



A bridge to rarefied-flow research

Temperature gradients and gas–surface interactions can drive small-scale gas transport.

Baier, Hardt, Shahabi and Roohi (2017): a radiometer-inspired Knudsen pump.

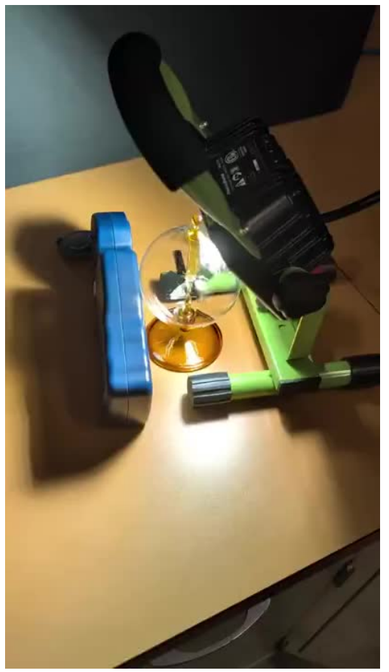
Lotfian and Roohi (2018): vane and ratchet geometries studied with DSMC.

The equipment already available

Sealed radiometer, LED lamp, lux meter and ruler.

Chronos camera and tripod; a phone can provide longer continuous video when needed.

Keep the source, camera, bulb and background positions in the run log.

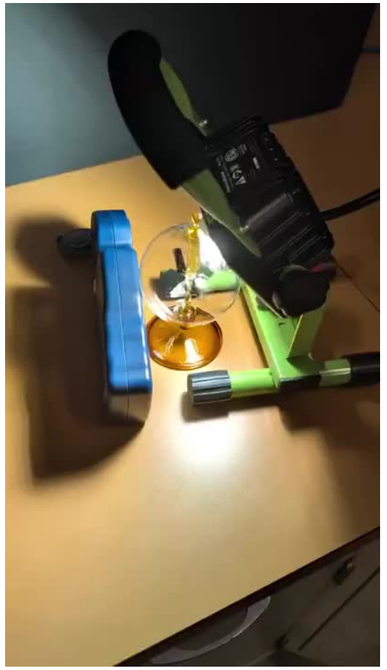


Prior recording: lamp and optical access

This is our prior apparatus, recorded in May 2026.

Identify the lamp, glass bulb, vanes and camera sightline.

Look for reflections and confirm a clear crossing location before collecting data.



Play course demonstration video

Prior recording: the vanes in motion

The film shows the actual rotor and a nearby lux meter.

Watch one identifiable appearance recur. Does it correspond to a full revolution?

The excerpt is for observation. Its playback time is not a verified physical acquisition time.



Play course demonstration video

The questions we can answer experimentally

At the same lamp setting, how does measured rotation change with illuminance?

Does speed remain constant, increase or decrease during a fixed observation window?

Does repeating a condition after warming reproduce its initial response?

What drives the vanes

The dark and bright faces absorb radiation differently. Their temperatures can differ.

Residual gas exchanges energy and momentum with the surfaces. Edge-region thermal effects create a net torque.

Losses oppose the motion. The usual rotation is thermally driven through the gas.

Where the Knudsen number enters

λ is molecular mean free path [m]. L is a chosen vane or gap length [m].

When molecular transport competes with the device length, gas–surface effects and edges matter.

The bulb is sealed. We cannot assign its Knudsen number without internal gas-state information.

$$\mathrm{Kn} = \lambda / L$$

Lux and radiant power

Lux describes light weighted by human visual sensitivity. Irradiance has units W/m^2.

Keep the same lamp spectrum, geometry and meter orientation. Compare matched-window mean rpm with measured lux.

Do not infer optical efficiency or assume an inverse-square law for a nearby extended lamp.

Heating makes this more than a speed-counting exercise

Illumination heats the vanes, bulb and nearby surfaces on different time scales.

The driving torque can change while the rotor is moving. A constant speed is not guaranteed.

Record initial state and compare early and late parts of the same window.

What the video measures

Each image has a frame number n. The capture rate is in frames per second.

Choose a reference frame n_o. Physical elapsed time t is in seconds.

A slow-motion file can play at 60 fps while the camera acquired at a much higher rate.

$$t = (n - n_0)/f_c$$

A passage is not always a revolution

Observe one complete turn and define the crossing you will count.

p is the verified number of selected events per revolution. A unique vane crossing gives p = 1.

Repeated black/bright appearances can create multiple events. Confirm p from the film.

From event counts to rotation speed

ΔN is the number of intervals between marked events. With nine marks, ΔN = 8.

Δt is elapsed physical time [s]. rpm is revolutions per minute.

Speed magnitude comes from timing. Rotation direction comes from watching the movie.

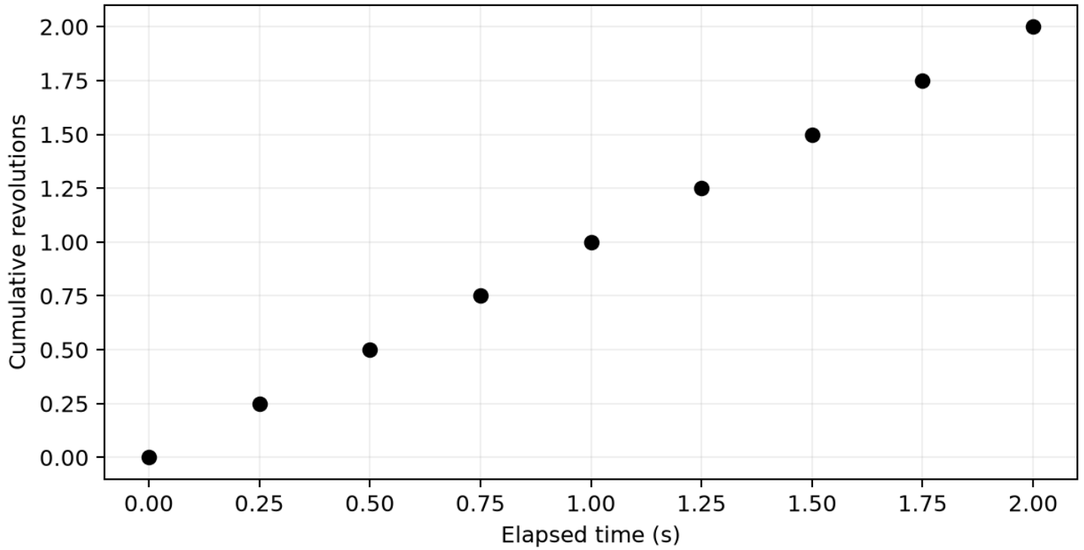
$$\text{rpm} = \frac{60\,\Delta N}{p\,\Delta t}$$

Worked timing example

Synthetic example: frames 0, 250, ..., 2000 at capture rate = 1000 fps.

Elapsed time is 2 s. If p = 4, eight event intervals represent two full turns.

Result: 60 rpm, or 2π rad/s. Using 60 playback fps would give 3.6 rpm.



Changing speed during start-up

Use consecutive original frame numbers for the same passage event. Each interval gives one average speed.

Place that speed at the interval midpoint. Do not label it instantaneous angular acceleration.

A long interval may indicate slowing or a missed event. Check the film.

$$\Delta t_i = \frac{n_{i+1} - n_i}{f_c}, \qquad \mathrm{rpm}_i = \frac{60}{p\,\Delta t_i}$$

Rotational dynamics

I is rotor inertia [kg·m²]. ω is angular speed [rad/s]. t is time [s].

The two torque terms represent the driving gas torque and resisting losses, both in N·m.

Newton’s rotational law relates acceleration to the difference between these torques.

$$I\frac{d\omega}{dt} = \tau_{\text{gas}} - \tau_{\text{loss}}$$

A simple transient approximation

Approximate loss torque as bω, with b in N·m·s. Assume constant driving torque.

Divide the torque balance by I. The angular acceleration plus (b/I) times angular speed equals the driving torque divided by I.

The mechanical time constant equals I/b. At steady rotation, the driving torque equals b times the final angular speed.

$$\tau_{\text{loss}} \approx b\omega, \qquad T_m = I/b$$

Solution of the previous ODE

Initial and final angular speeds use subscripts 0 and ∞ [rad/s]. Subtract the steady solution and integrate the homogeneous remainder.

The remaining departure from steady speed decays exponentially.

From rest, the speed reaches about 63% of its final value after one mechanical time constant.

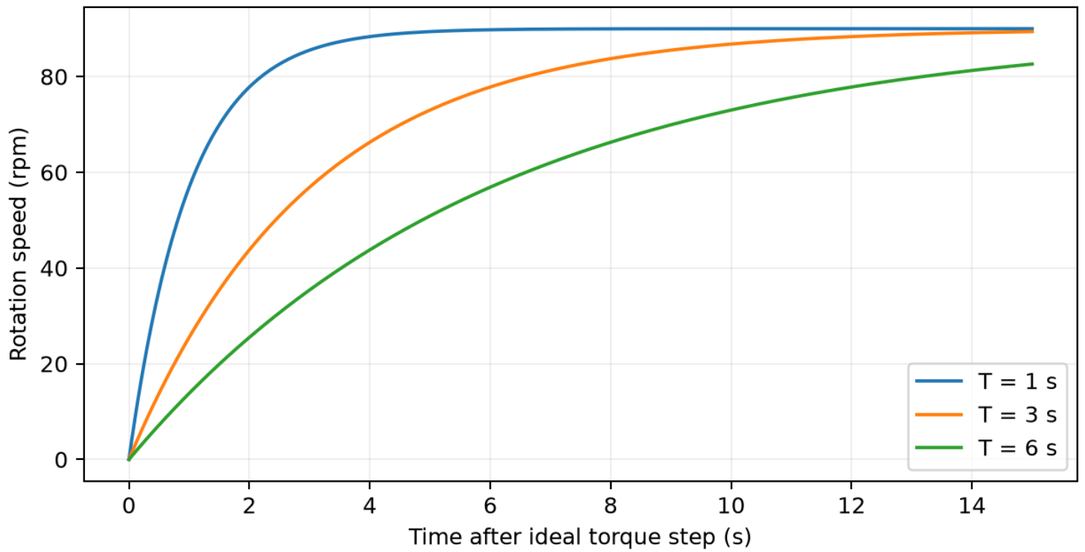
$$\omega(t) = \omega_\infty + (\omega_0 - \omega_\infty)e^{-t/T_m}$$

Heating can change the response

The illustrated curves assume an ideal constant torque step and final speed of 90 rpm.

Larger T gives slower response. These curves are synthetic examples.

In the bulb, thermal inertia changes vane temperatures and torque. An observed time constant can combine thermal and mechanical effects.

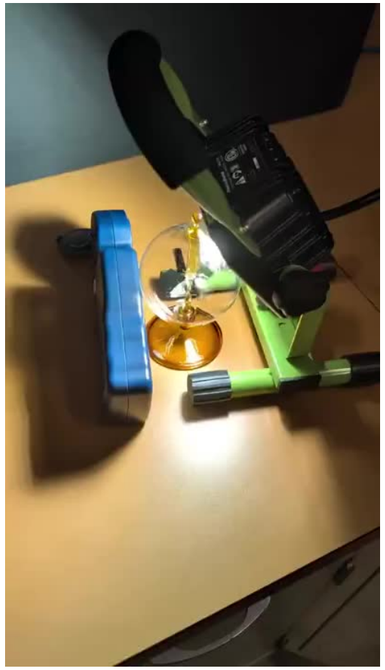


Bench step 1: assemble and measure

Set the bulb upright, lamp on a stable support and camera on a tripod. Keep the background out of the strongest beam.

Measure lux beside the rotor plane, at a repeatable position facing the lamp. Record distance and lamp setting.

Lock exposure, gain and focus. Show the whole rotor, but make a repeatable passage easy to see.



Bench step 2: prove that the movie is usable

Record a short preview. Read actual capture fps from the camera settings and save that setting with the run.

Watch a complete turn and verify the event count per revolution. Check several consecutive crossings frame by frame.

For slow rotation, favor enough physical duration over maximum fps. Check available camera memory before the full run.

Bench step 3: define the measurement window

Start recording before switching on the lamp. Record the illumination-on frame as the physical time origin.

Use a common 60–90 s window for the comparison, if enough crossings occur. Aim for at least six verified intervals.

If a condition is too slow, extend the common window or reduce the number of conditions. Do not invent missing crossings.

Bench step 4: collect A, B, C and repeat A

A: reference illuminance. B and C: two different safe distances, using the same lamp setting. Measure actual lux each time.

Begin A, B and C from comparable initial rotor and thermal states. Log cooling time and whether the rotor has stopped.

Repeat A after cooling as a repeatability check. Save one continuous movie and one log row per run.

Bench step 5: test thermal memory if time permits

After a reference illumination run, block the light without touching the bulb and continue recording.

Observe whether the rotor keeps moving, slows or changes direction. Record when the light was blocked.

A warm return to A can be compared with a cooled A, but label it as a thermal-history test.

From a movie to defensible passage data

Run Mark_Lab6_Passage_Frames.m. Arrows navigate; Space marks; Backspace undoes; Q saves.

Always mark the same event in the same direction. Use a one-frame step at each crossing.

Run Run_Lab6_Radiometer.m with actual capture fps and verified event count. Review flagged intervals in the movie.

Comparing the runs in MATLAB

Fill Lab6_Comparison_Log.csv with each passage file, capture fps, verified event count, lux and lamp-on frame.

Enter the same requested time window and label the initial state. Run Compare_Lab6_Illumination_Runs.m.

The outputs show interval rpm versus physical time, mean rpm versus lux, early/late means and actual covered times.

What the report should show

A labeled setup photo and the recorded settings. One run with cumulative revolutions and interval rpm.

Across runs: matched-window mean rpm versus measured lux, showing repeats and initial state.

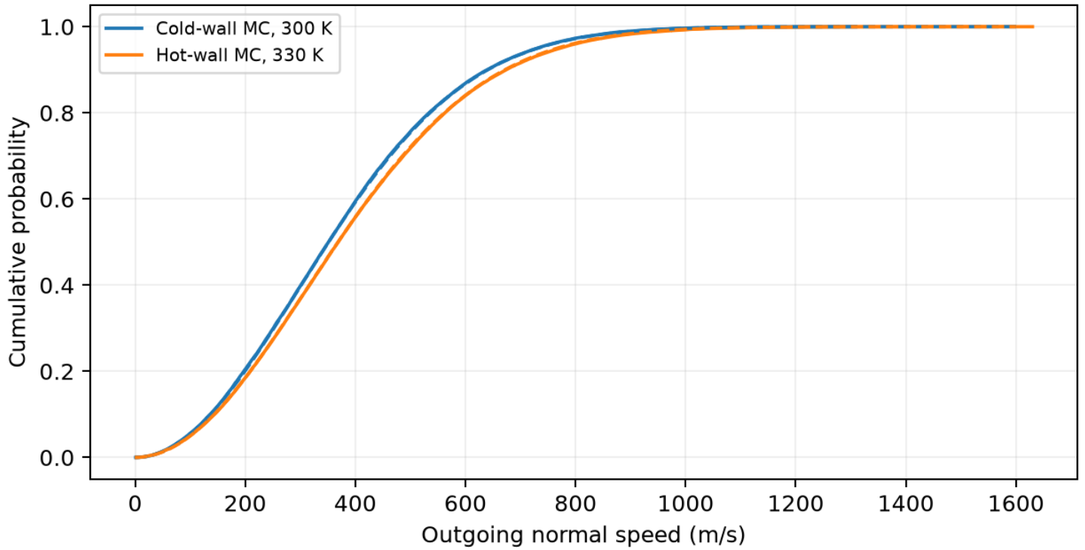
Explain early/late differences, repeatability and limits of the constant-torque approximation.

Molecular Monte Carlo: the physical idea

Sample molecules striking two faces at different temperatures.

Each collision transfers momentum. Average many events to estimate a pressure difference.

A full rotor model also needs edges, gas collisions, confinement and heat transfer.



The Monte Carlo algorithm

Replace a difficult average by many samples from the physical collision distribution.



The random seed makes a realization reproducible. More samples reduce sampling scatter; the physical assumptions remain the same.

Why wall-crossing velocities need special sampling

T is temperature [K]. Molecular mass is m (subscript g) [kg]; Boltzmann’s constant is k (subscript B) [J/K].

Faster normal molecules cross a wall more often. The collision sample is flux-weighted, not an ordinary bulk Gaussian sample.

For a diffuse reflection, draw U uniformly between zero and one. The equation gives outgoing normal speed v (subscript n) [m/s].

$$v_n = \sqrt{-\frac{2k_B T}{m_g} \ln U}, \qquad v_n > 0$$

From molecular impulse to mean wall force

Each collision transfers normal impulse j (subscript n) [kg m/s], from the incident and outgoing normal speed magnitudes [m/s].

Γ is the number of incident molecules per area per time [m^{-2} s^{-1}]. Wall pressure P [Pa] is incident flux times mean impulse per collision.

Compare the hotter and colder faces under the same ideal incident reservoir. Define positive force from hot toward cold.

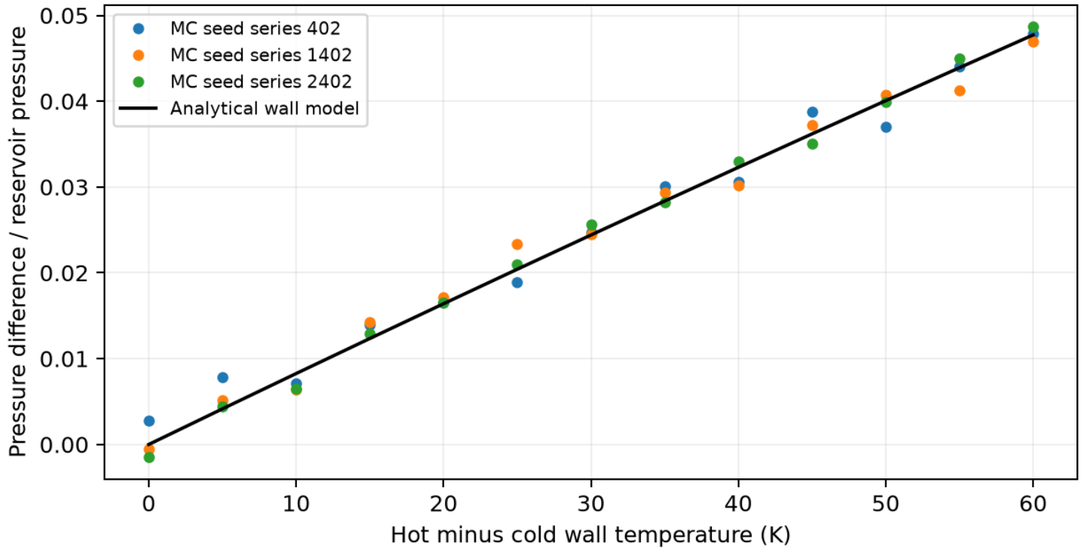
$$j_n = m_g(v_{\text{in}} + v_{\text{out}}), \qquad P_{\text{wall}} = \Gamma \langle j_n \rangle$$

The analytical check for the molecular activity

T is temperature [K]; subscripts g, h and c mean reservoir, hot wall and cold wall.

P is pressure (subscript g: reservoir). a is diffuse-reflection fraction, not optical absorptivity.

ΔP is hot-face minus cold-face pressure. Compare MC with the analytical expectation below.



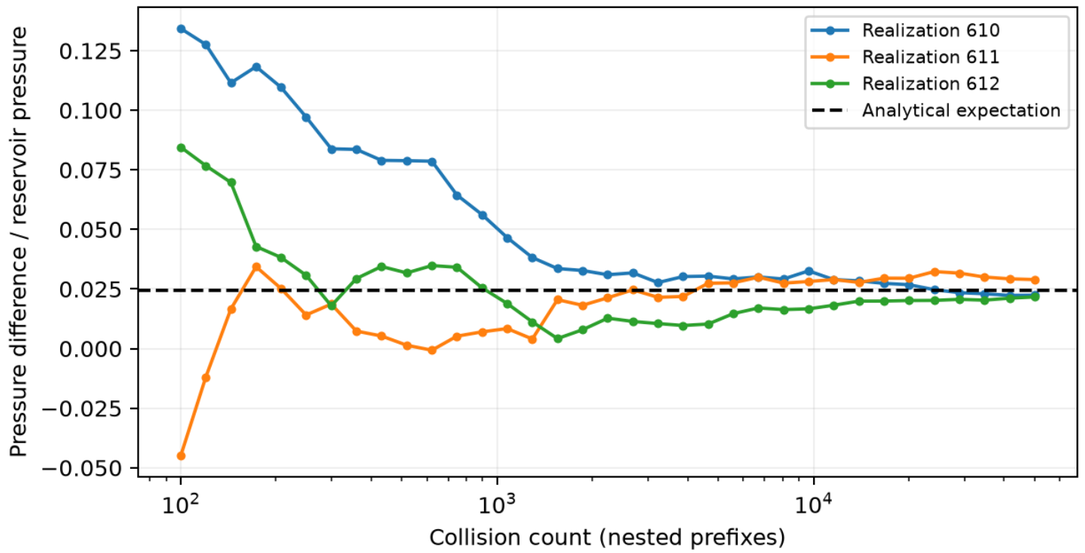
$$\frac{\Delta P}{P_g} = \frac{a}{2}\left(\sqrt{\frac{T_h}{T_g}} - \sqrt{\frac{T_c}{T_g}}\right)$$

Why the number of molecular samples matters

N is the number of sampled collision events. Each plotted realization uses nested prefix averages.

Independent events with finite variance give sampling error that decreases approximately as one over the square root of N.

About four times as many samples are needed to halve that error. More samples do not supply missing edge or gas-collision physics.



Required activity in the Pre-Lab 6 notebook

Run Q6 in the same Colab notebook as Q1–Q5. It explains the algorithm and implements the molecular sampling.

Submit the velocity-distribution and force-versus-temperature figures. Discuss equal temperatures, reversal, reflection fraction and sample count.

Run the two thermal-state examples. Explain why warmer faces can have a smaller temperature contrast and weaker driving force.

Connecting the molecular activity to your measured runs

In the report, relate gas recoil to the observed rotation direction and measured rpm-versus-lux trend.

Compare early/late speeds and cooled repeats. Discuss whether thermal contrast could evolve while the lamp remains on.

Pressure, vane temperatures, enclosure effects and losses are not all known. Use the wall model to interpret mechanisms, not as a calibrated rpm curve.

Before leaving the laboratory

Show a usable movie, actual capture fps, verified event count and at least one accepted passage table.

Show one rpm trace and the comparison of the illumination runs. Save all raw movies and the completed log.

Lab 6 report: Monday, December 7, 11:59 PM. Include only measured quantities and theory used in the discussion.